

Esperimento 2 - confronto setup LSTM

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 Date	June 2, 2026
 Select	Esperimenti

Confronto setup LSTM

In questo esperimento sono stati confrontati diversi setup ATS sotto attacco. In particolare:

- LSTM trainato su dati di 13 anni (baseline paper)
- LSTM trainato su dati di 3 anni
- LSTM trainato ogni 3 mesi in modo cumulativo
- LSTM trainato ogni 3 mesi con rolling window di 5 anni

Valutazione modelli

Vediamo innanzitutto le performance dei vari modelli.

Parametri per valutazione:

```
simulation_start_date = '2021-01-01'  
simulation_end_date = '2023-10-23'  
  
sequence_length = 50  
  
quarterly_data_start_date = '2013-01-01'  
data_end_date = '2023-10-23'
```

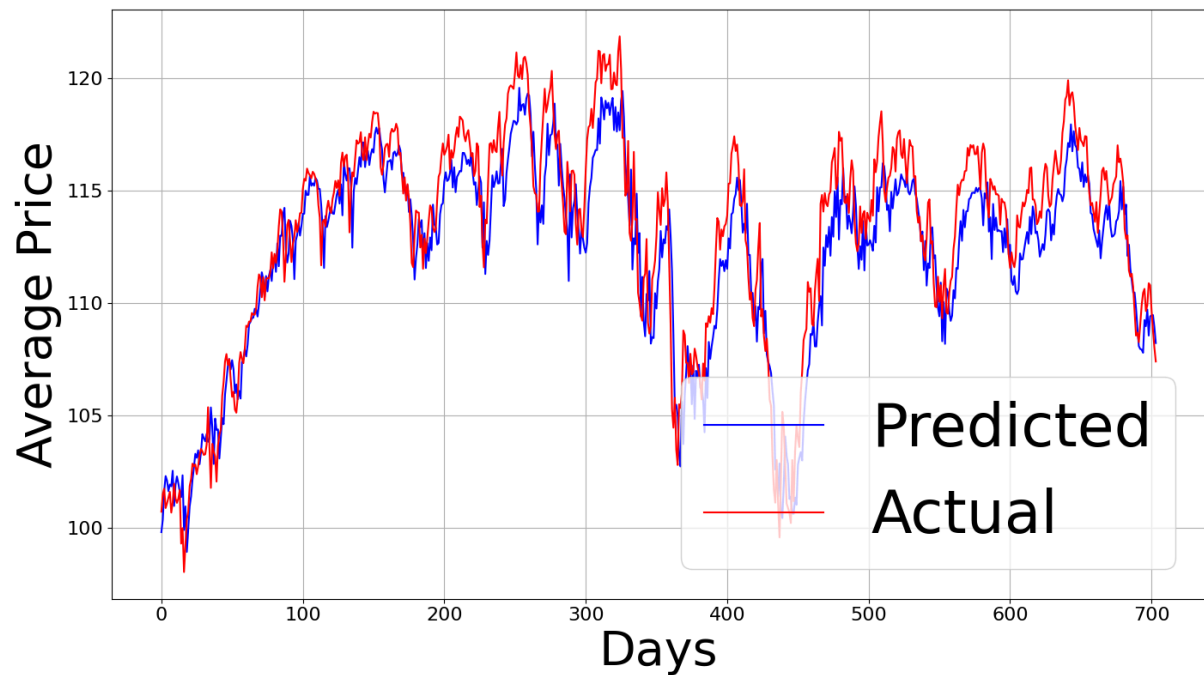
LSTM 13 years

Addestrato su :

```
start_date = '2010-01-01'  
end_date = '2023-10-23'  
sequence_length = 50
```

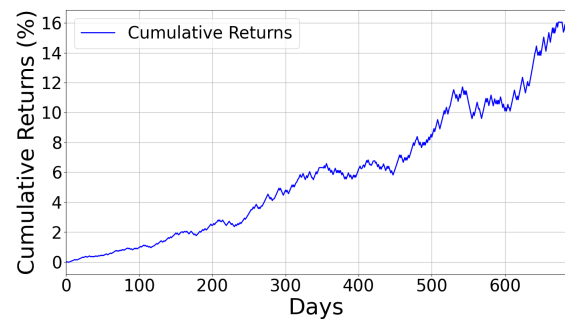
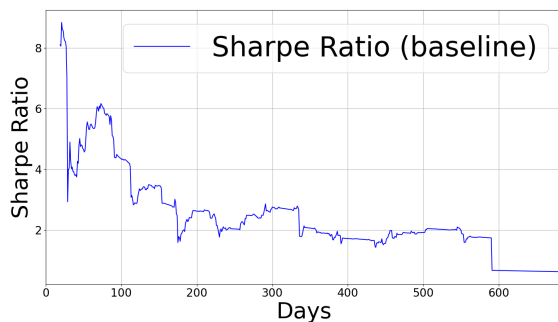
Accuratezza (model_analysis.ipynb):

RMSE medio $\approx$ 4.9121



Profittabilità (ats.ipynb):

- trading strategy = Moving Avarage



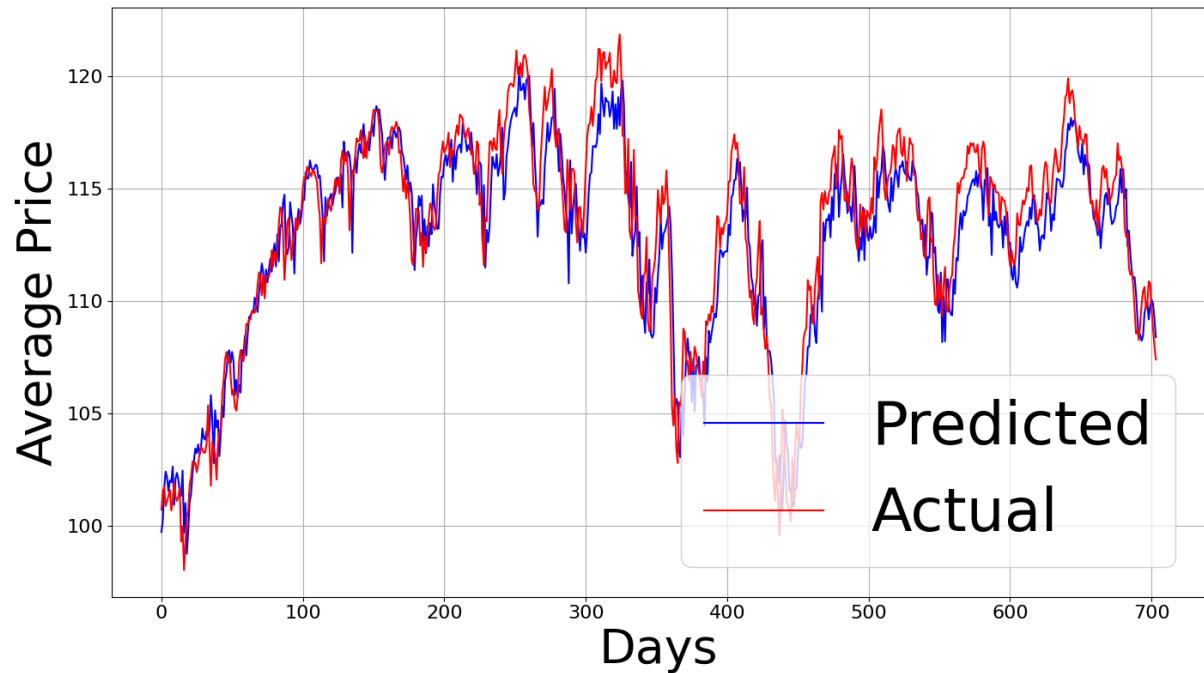
LSTM 3 years

Addestrato su:

```
start_date = '2020-01-01'  
end_date = '2023-10-23'  
sequence_length = 50
```

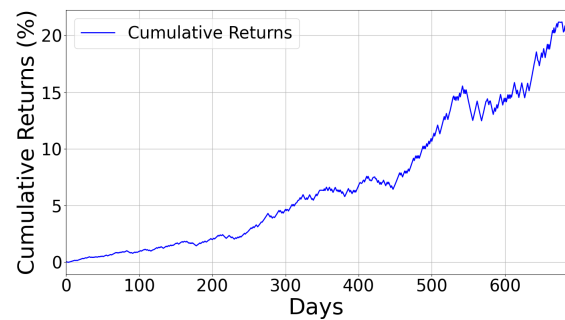
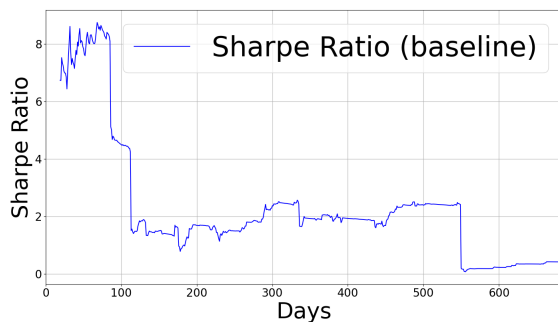
Accuratezza (model_analysis.ipynb):

RMSE medio ≈ 4.9632



Profittabilità (ats.ipynb):

- trading strategy = Moving Avarage



LSTM cumulative retrain

Addestrato su:

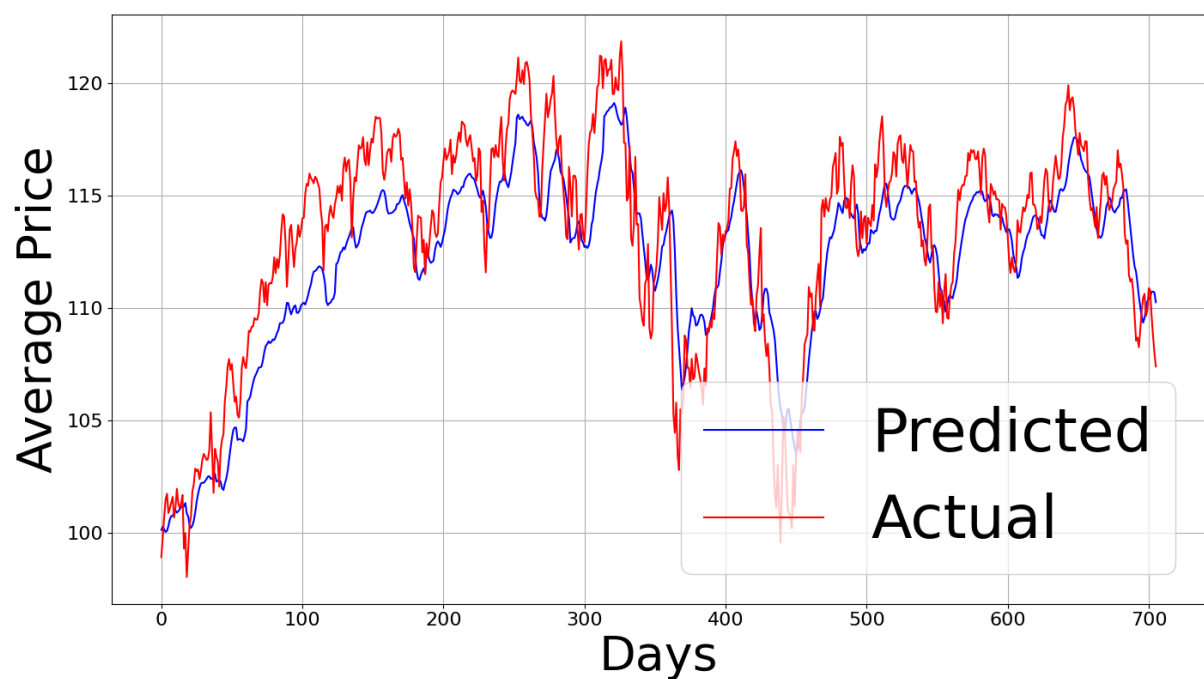
```

simulation_start = '2020-01-01'
simulation_end = '2023-10-23'
sequence_length = 50
data_start = '2013-01-01'
data_end = '2023-10-23'
months = 3

```

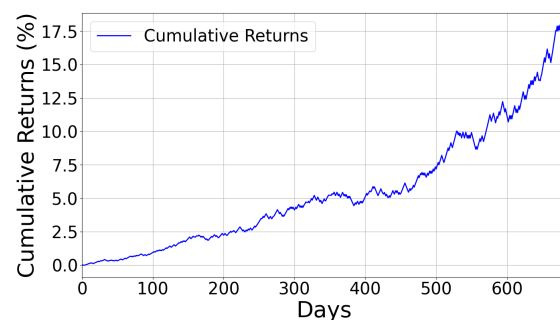
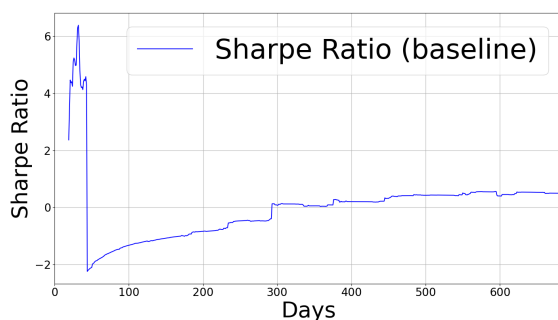
Accuratezza (model_analysis.ipynb):

RMSE $\approx$ 7.0087



Profittabilità (ats.ipynb):

- trading strategy = Moving Avarage



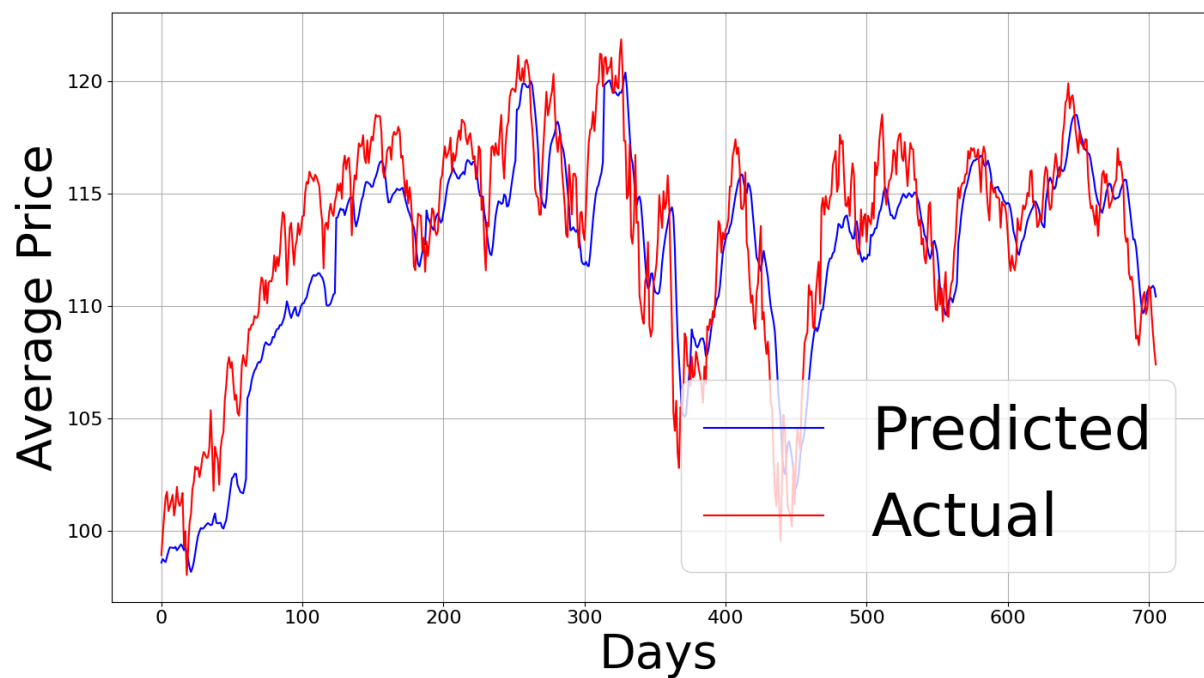
LSTM rolling window retrain

Addestrato su:

```
simulation_start = '2020-01-01'  
simulation_end = '2023-10-23'  
sequence_length = 50  
data_start = '2020-01-01'  
data_end = '2023-10-23'  
months = 3  
lookback_years = 3
```

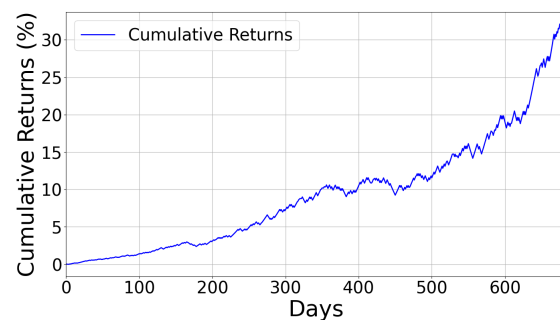
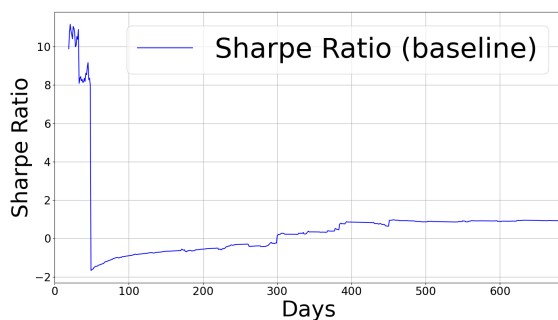
Accuratezza (model_analysis.ipynb):

RMSE $\approx$ 4.9681



Profittabilità (ats.ipynb):

- trading strategy = Moving Avarage



Attacco

L'attacco si svolge esattamente come nell'esperimento precedente (esperimento 1). Per ogni setup vengono innanzitutto calcolati i giorni di attacco in base a 6 diverse timing policies:

- uniform
- high_volatility
- high_pred_volatility
- strong_trend
- weak_trend
- near_strategy_boundary

Per ogni setup e timing policy viene poi simulato un attacco per ogni giorno individuato introducendo la minima perturbazione che altera il risultato baseline.

L'efficacia di un attacco su un determinato setup viene calcolata tramite il `nontrivial_attack_success_rate`.

Parametri attacco

```
SEQUENCE_LENGTH = 50
N_ATTACK_DAYS   = 12
MAX_ABS_DELTA   = 0.35
GRID_STEPS      = 35

QUARTERLY_SIMULATION_START = "2021-01-01"
QUARTERLY_SIMULATION_END   = "2023-10-23"
QUARTERLY_DATA_START       = "2013-01-01"
QUARTERLY_MONTHS           = 3

ALL_TICKERS = [
    "GOOGL", "AMZN", "AAPL", "PEP", "JNJ", "PFE", "MRK", "A
    BBV", "PG", "KO",
    "WMT", "JPM", "BAC", "GS", "V", "XOM", "CVX", "COP", "B
    P", "BA",
    "MMM", "HON", "GE", "T", "VZ", "TMUS", "HSY", "DUK", "S
```

```
0", "EXC", "AEP",
    "AMT", "PLD", "SPG", "BHP", "RIO", "VALE", "FCX",
]
```

Risultati

Moving Avarage

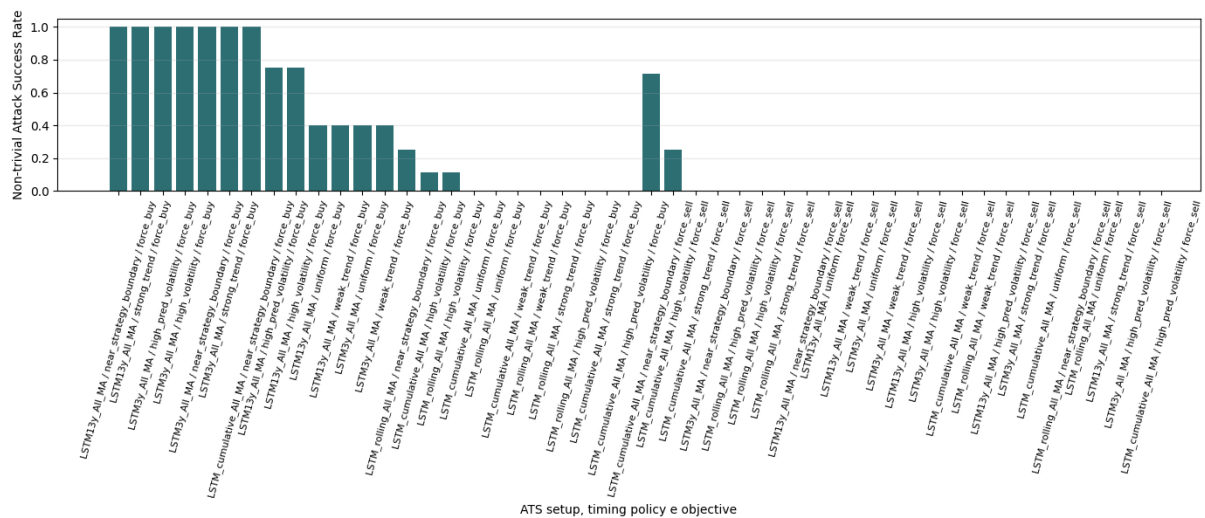
Setup summary

setup	trials	attack_success_rate	baseline
_target_rate	nontrivial_trials	nontrivial_successes	mean
_min_abs_delta_norm	median_min_abs_delta_norm	mean_delta_	
final_cr	nontrivial_attack_success_rate		
LSTM3y_All_MA	144	0.638889	
0.500000	72	20	
0.003804		0.0	0.028398
0.277778			
LSTM13y_All_MA	144	0.611111	
0.465278	77	21	
0.006023		0.0	-0.024001
0.272727			
LSTM_cumulative_All_MA	144	0.576389	
0.465278	77	16	
0.017831		0.0	0.005784
0.207792			
LSTM_rolling_All_MA	144	0.500000	
0.486111	74	2	
0.000278		0.0	0.000000
0.027027			

Policy summary

timing_policy	trials	attack_success_rate	baseli
ne_target_rate	nontrivial_trials	mean_delta_cr	nontrivia
l_successes	mean_min_abs_delta_norm	nontrivial_attack_suc	cess_rate
near_strategy_boundary	96	0.739583	

0.416667	56	0.004516
31	0.022113	0.553571
high_volatility	96	0.583333
0.479167	50	0.032537
10	0.002857	0.200000
strong_trend	96	0.552083
0.489583	49	0.029455
6	0.010000	0.122449
uniform	96	0.541667
0.500000	48	-0.032364
4	0.000769	0.083333
weak_trend	96	0.541667
0.500000	48	-0.022949
4	0.000769	0.083333
high_pred_volatility	96	0.531250
0.489583	49	0.003003
4	0.000784	0.081633



Rate of Change

Setup summary

setup	trials	attack_success_rate	baseline
_target_rate	nontrivial_trials	nontrivial_successes	mean
_min_abs_delta_norm	median_min_abs_delta_norm	mean_delta	
final_cr	nontrivial_attack_success_rate		
LSTM3y_All_ROC	144	0.736111	

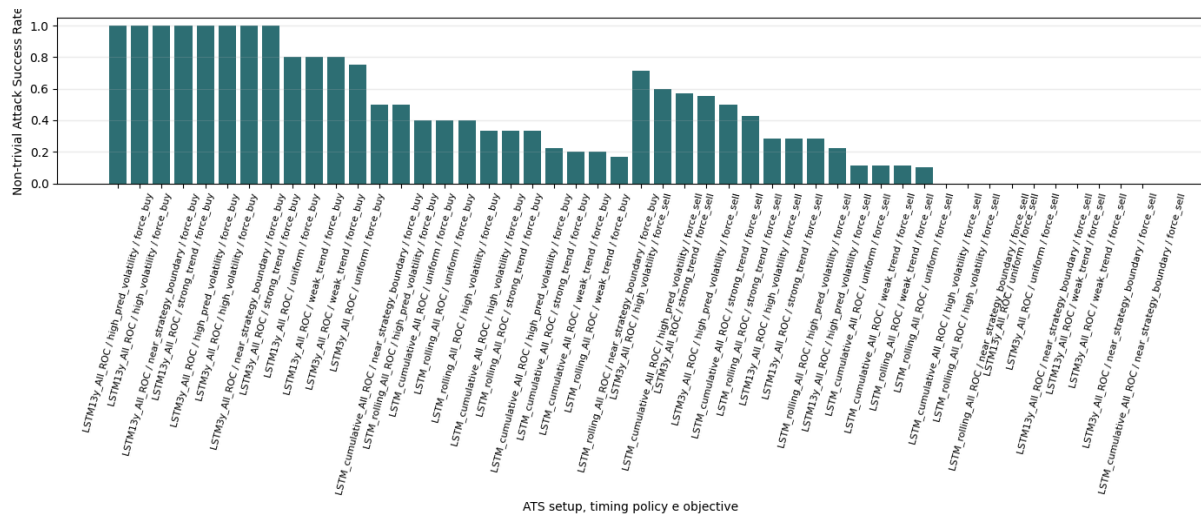
0.458333	78		40
0.031792		0.0	-0.089780
0.512821			
LSTM13y_All_ROC	144		0.687500
0.423611	83		38
0.026869		0.0	-0.210393
0.457831			
LSTM_cumulative_All_ROC	144		0.527778
0.368056	91		23
0.038421		0.0	0.056639
0.252747			
LSTM_rolling_All_ROC	144		0.416667
0.402778	86		2
0.004833		0.0	0.0
0.023256			

Policy summary

Nota: risultati leggermente sminchiati perché attacco simulato con configurazione per LSTM_rolling impostata a cumulative. Verificare

	timing_policy	trials	attack_success_rate	baseline_target_rate	nontrivial_trials	mean_delta_cr	nontrivial_successes	mean_min_abs_delta_norm	nontrivial_attack_success_rate
strong_trend		96	0.708333						
0.416667		56	-0.054977						
28		0.056765					0.500000		
high_pred_volatility		96	0.677083						
0.364583		61	-0.043132						
30		0.054000					0.491803		
high_volatility		96	0.677083						
0.375000		60	-0.273121						
29		0.033538					0.483333		
uniform		96	0.572917						
0.437500		54	-0.035322						
13		0.014364					0.240741		
weak_trend		96	0.541667						
0.416667		56	0.065266						

12	0.021538	0.214286
near_strategy_boundary	96	0.552083
0.437500	54	0.026789
11	0.013585	0.203704



Bollinger Bands

Setup summary

	setup	trials	attack_success_rate	baseline_target_rate	nontrivial_trials	nontrivial_successes	median_min_abs_delta_norm	mean_delta_final_cr	nontrivial_attack_success_rate
LSTM_rolling_All_BB		144	0.284722						
		103							
		0.0							
		0.0							
LSTM3y_All_BB		144	0.118056						
		127							
		0.0							
		0.0							
LSTM_cumulative_All_BB		144	0.104167						
		129							
		0.0							
		0.0							
LSTM13y_All_BB		144	0.076389						
		133							

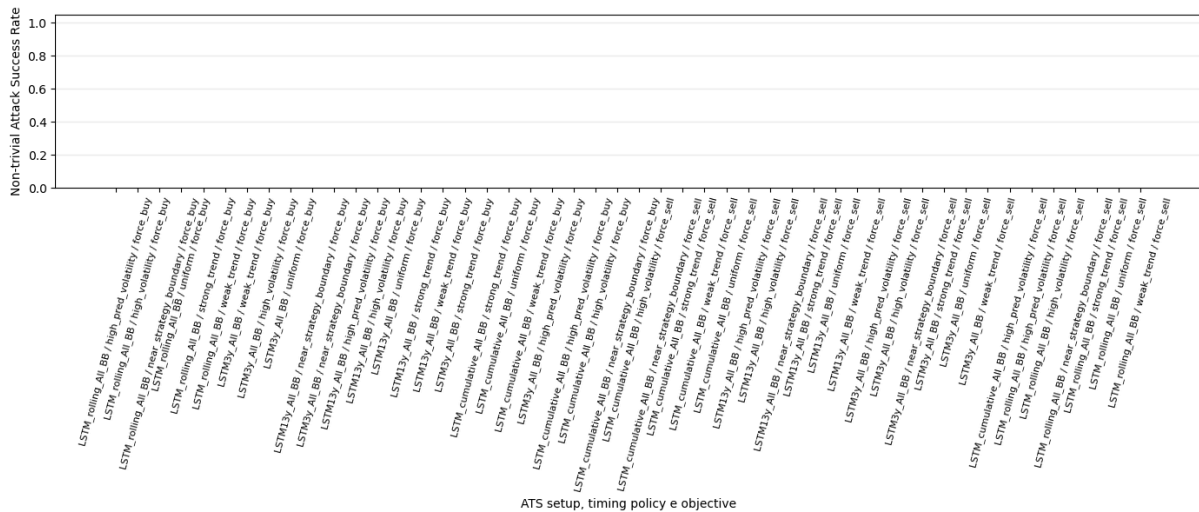
0.0
0.0

0.0

0.0

Policy summary

	timing_policy	trials	attack_success_rate	baseline_target_rate
nontrivial_trials				
nontrivial_successes				
mean_min_abs_delta_norm				
nontrivial_attack_success_rate				
high_pred_volatility	96	0.104167		
0.104167	86	0		
0.0		0.0		
high_volatility	96	0.166667		
0.166667	80	0		
0.0		0.0		
near_strategy_boundary	96	0.166667		
0.166667	80	0		
0.0		0.0		
strong_trend	96	0.125000		
0.125000	84	0		
0.0		0.0		
uniform	96	0.156250		
0.156250	81	0		
0.0		0.0		
weak_trend	96	0.156250		
0.156250	81	0		
0.0		0.0		



Differenze rispetto a esperimento precedente

- Il preprocessing degli LSTM conteneva un leakage che faceva il fitting per lo scaler su tutti i dati, non solo dati di train.
- I giorni di inizio e fine della simulazione vengono passati a mano per assicurarsi che siano gli stessi per ogni modello (che utilizzerebbe preprocessing diversi)